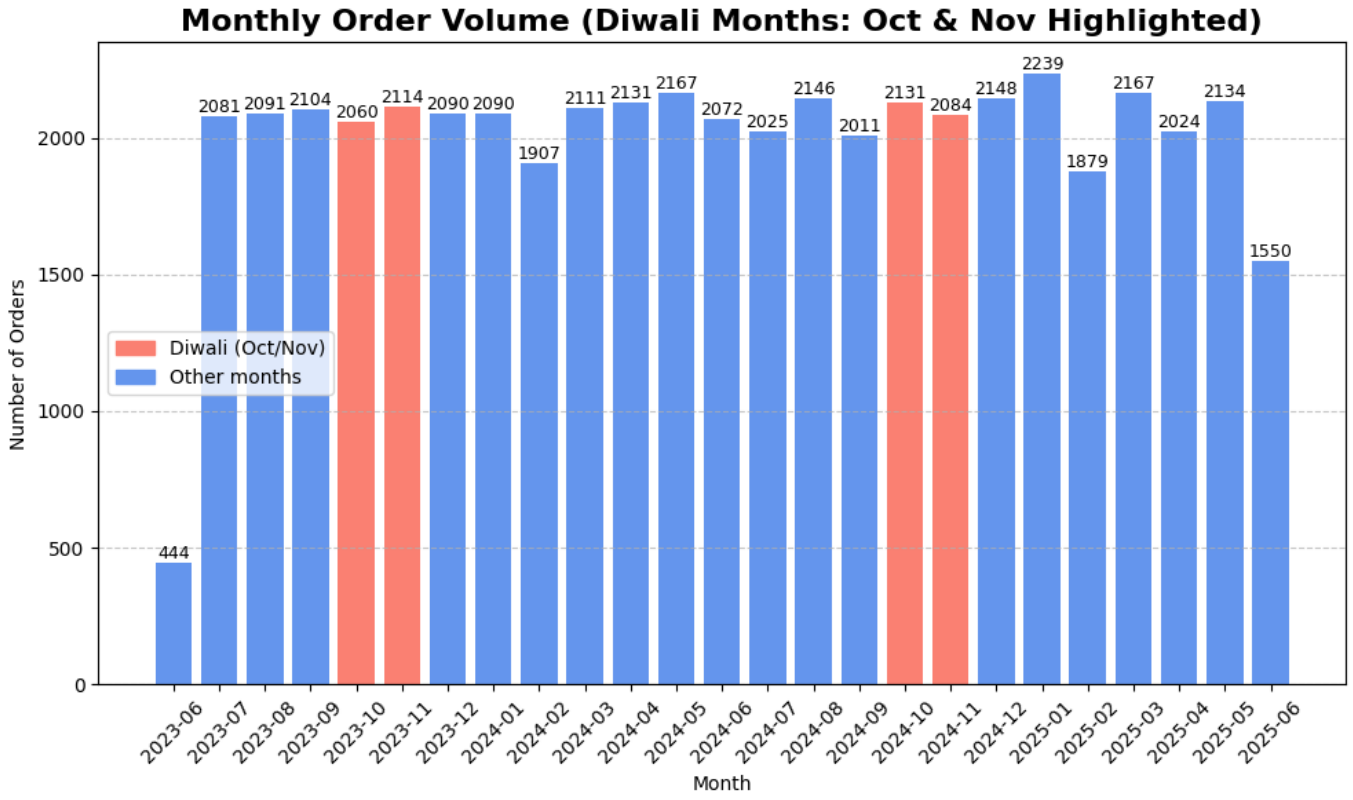
Data Analyst Intern: Bindiya Anand (Inter563)

IndiaKart E-Commerce

Phase 2: Exploratory Data Analysis

Kindly view the Python Notebook file in this link: [Python Notebook file link]

1. Month Order Value

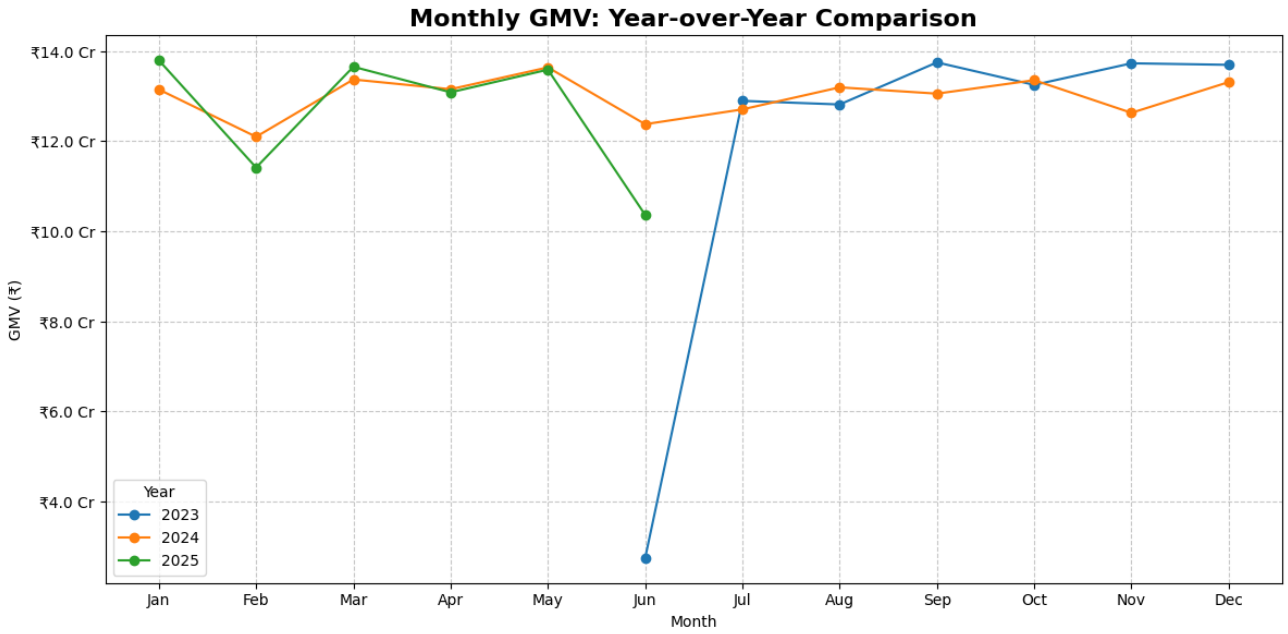


Business question: *How does order demand change over time, and is there a visible Diwali-season pattern?*

The monthly order volume is relatively stable across most of the observed period, generally remaining around 1,900-2,200 orders per month. January 2025 records the highest monthly order count at 2,239, while October and November do not consistently emerge as the highest-volume months in either year, so a strong Diwali-related order spike is not evident from these results. June 2023 and June 2025 have lower counts, but these may represent partial months and should not be compared directly with complete months.

- **Business implication:** IndiaKart should use its actual historical demand patterns to plan inventory and campaigns rather than assume that Diwali automatically produces the largest order volume.

2. Monthly Revenue (GMV), Year-over-Year

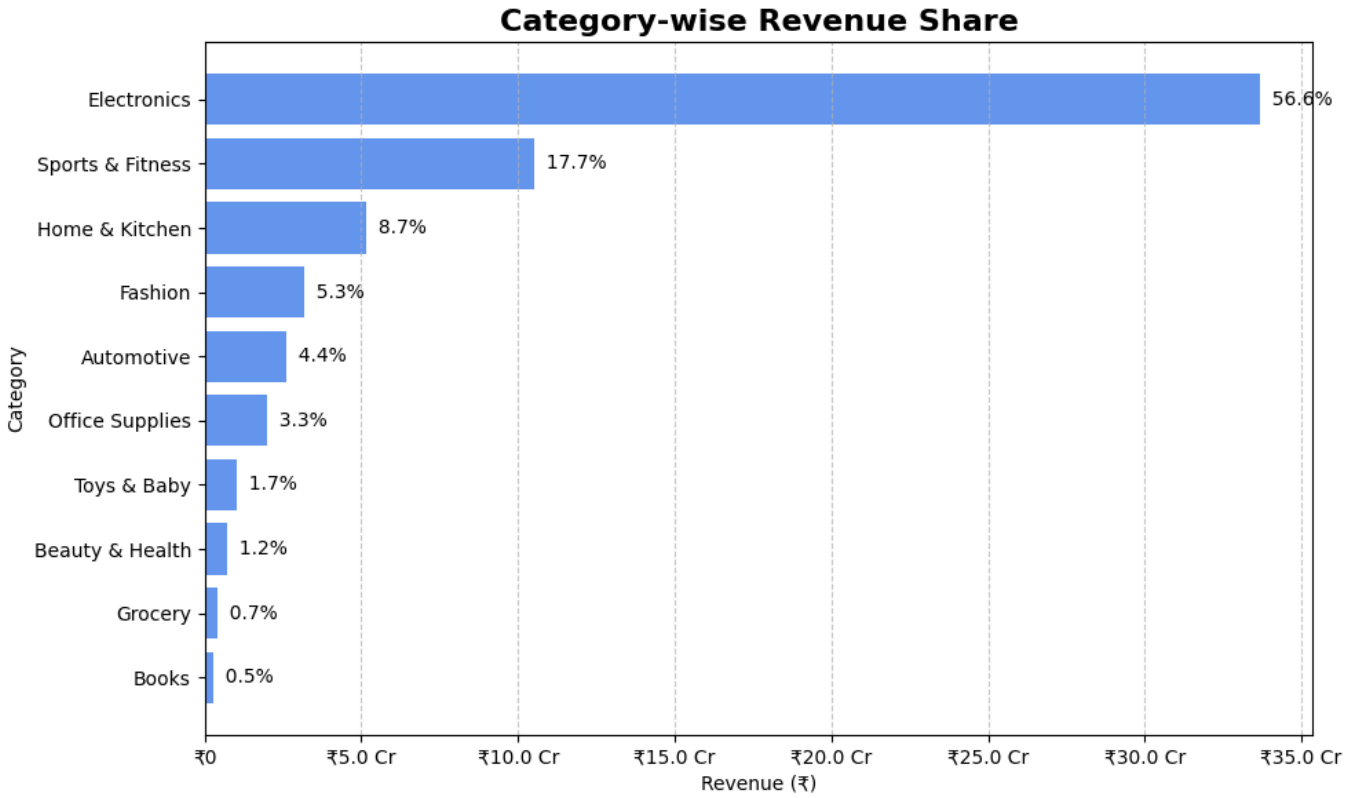


Business question: *How does the total value of orders change over time, and how does performance compare across years?*

Monthly GMV is generally around ₹12-₹14 crore during the observed full months, with January 2025 recording the highest value in the notebook output at approximately ₹13.79 crore. The year-over-year comparison shows mixed performance: January and March 2025 exceed the corresponding 2024 values, while February, April and May 2025 are lower; June also shows a decline, although the comparison may be affected by incomplete-month data. Because this chart includes all order statuses, it represents GMV as defined in the PRD, not revenue from delivered orders alone.

- **Business implication:** IndiaKart should investigate which products, order values and customer behaviours explain the month-to-month changes in GMV before deciding whether to increase promotions or adjust pricing.

3. Category-wise Revenue Share

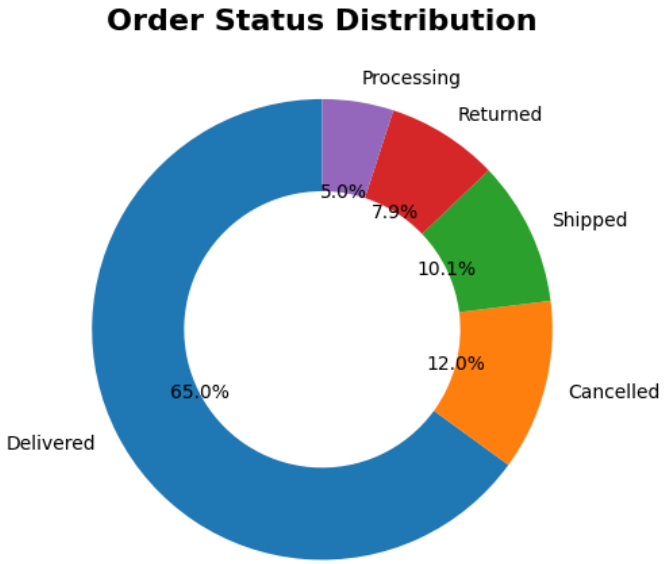


Business question: *Which product categories contribute the most revenue?*

Electronics is the leading revenue category, contributing approximately 56.6% of the total category revenue, followed by Sports & Fitness at 17.7%. Together, these two categories contribute approximately 74.2% of the revenue represented in the category-level analysis, while Books, Grocery and Beauty & Health contribute relatively small shares individually.

- **Business implication:** Electronics is a major revenue driver, but this concentration may create business risk if demand weakens in that category. IndiaKart could investigate opportunities to grow smaller categories while continuing to support its strongest performers.

4. Order Status Distribution

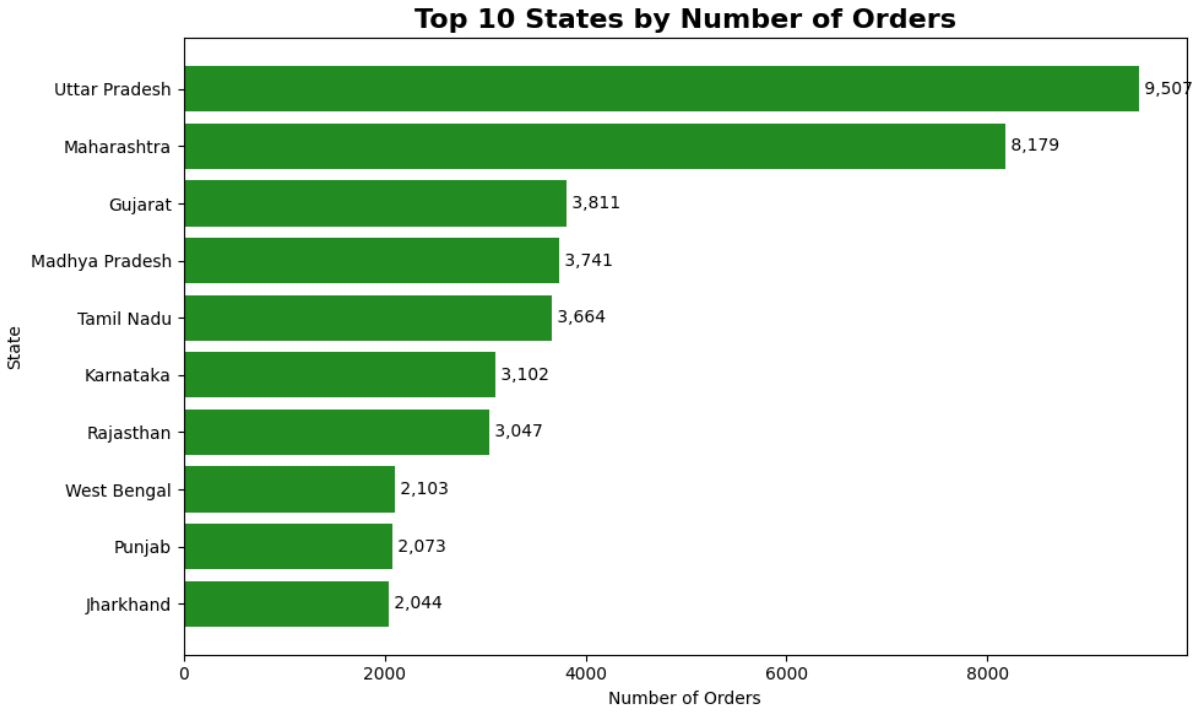


Business question: *What proportion of orders are delivered, cancelled, returned or still in progress?*

Of the 50,000 orders, 64.998% are Delivered, making it the dominant order status. Cancelled orders account for approximately 12.0%, Returned orders for 7.9%, Shipped orders for 10.1% and Processing orders for 5.0%; these results indicate that a substantial share of orders are either cancelled, returned or not yet marked as delivered.

- **Business implication:** IndiaKart should investigate the reasons behind cancellations and returns and monitor the time taken for shipped and processing orders to reach delivery. The status distribution alone does not establish why these orders are not delivered.

5. Top 10 States by Order

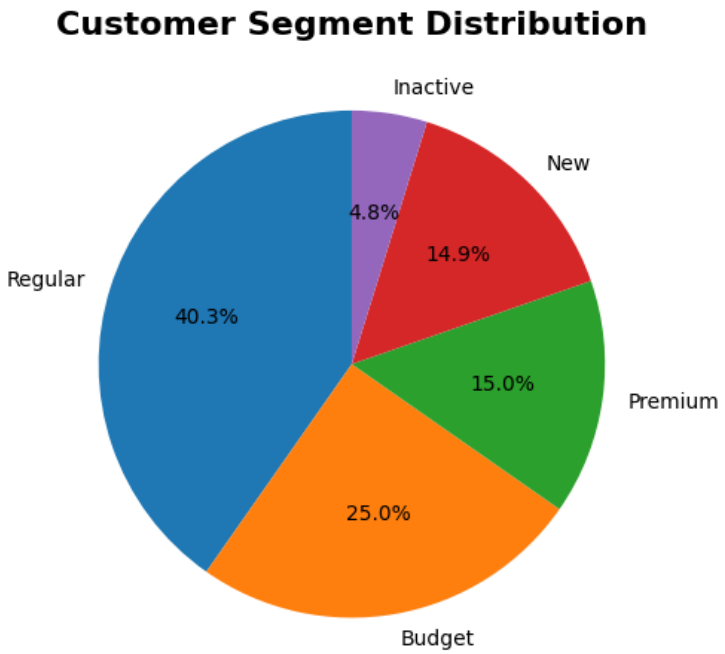


Business question: *Which states generate the highest order volumes?*

Uttar Pradesh leads with 9,507 orders, followed by Maharashtra with 8,179 orders. Gujarat, Madhya Pradesh and Tamil Nadu follow, while the remaining states in the top 10 have lower order volumes; Uttar Pradesh and Maharashtra together account for approximately 35.4% of all orders.

- **Business implication:** IndiaKart could prioritize delivery capacity, customer support and targeted marketing in Uttar Pradesh and Maharashtra while examining the potential for growth in states with lower order volumes. Order volume alone does not establish profitability or customer value by state.

6. Customer Segment Distribution

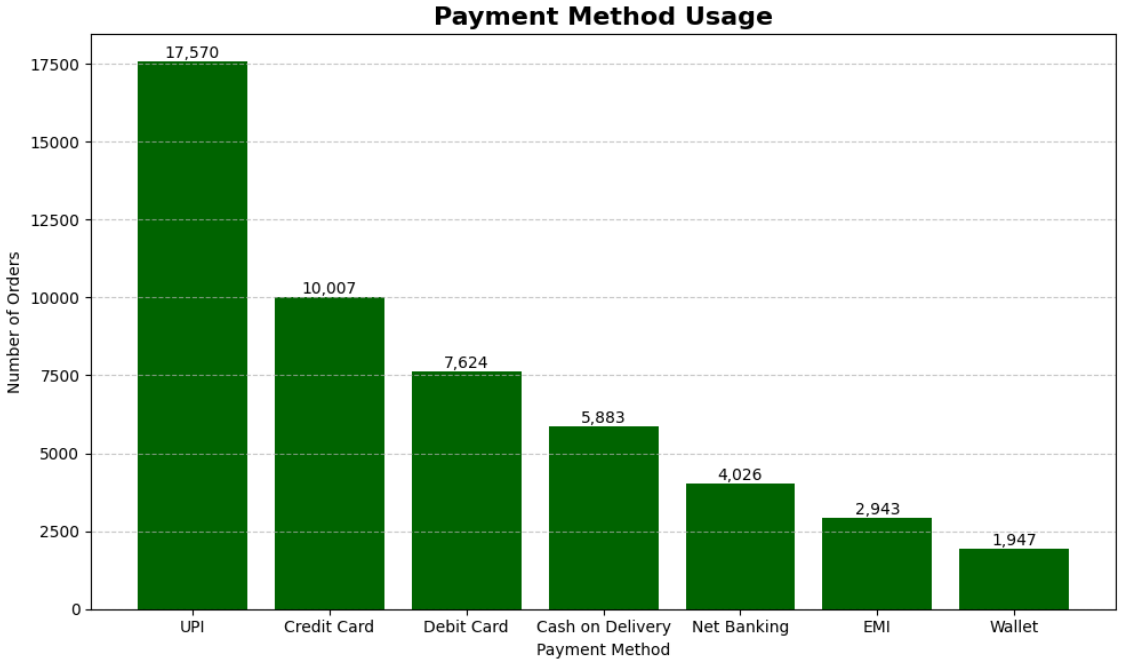


Business question: *How is the customer base distributed across segments?*

Regular customers form the largest segment, with 4,026 customers (40.26%), followed by Budget customers with 2,505 (25.05%) and Premium customers with 1,501 (15.01%). New customers account for 14.88%, while Inactive customers account for 4.80%, indicating that the customer base includes both established customers and a meaningful group of newer customers.

- **Business implication:** IndiaKart could focus on retaining Regular customers, encouraging suitable Budget customers to make repeat purchases and helping New customers complete their next purchase. The segment labels should not be treated as proof of spending behaviour without further analysis.

7. Payment Method Usage

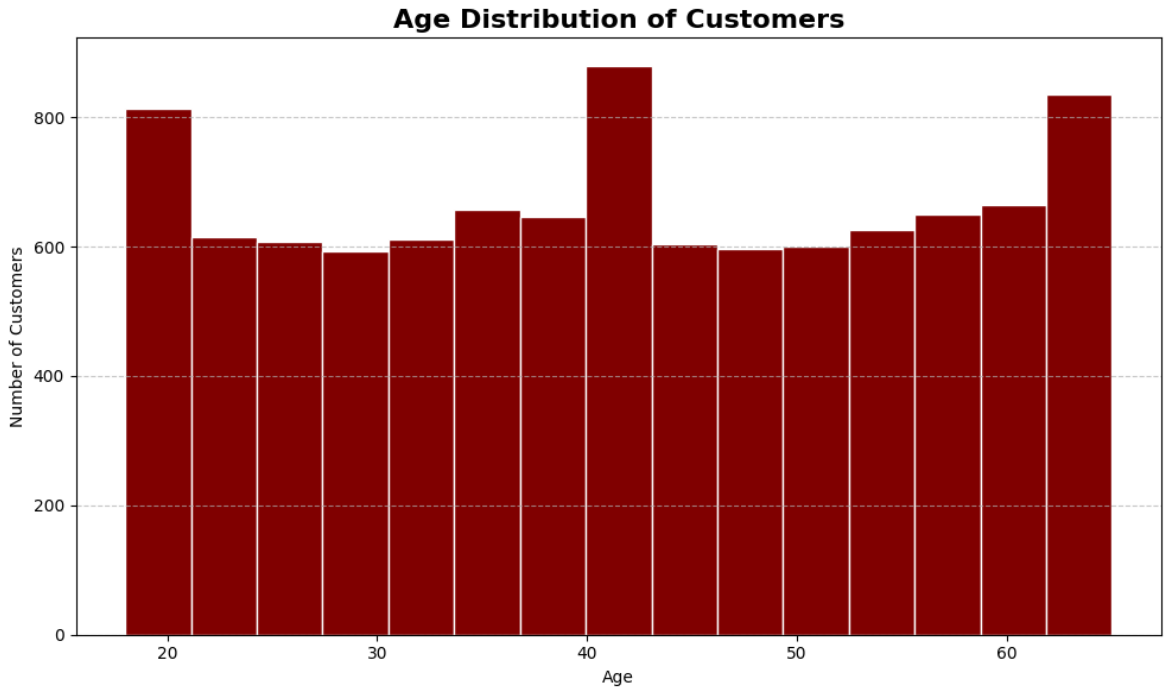


Business question: *Which payment methods are most commonly used for orders?*

UPI is the most frequently used payment method, accounting for 17,570 orders (35.14%), followed by Credit Card with 10,007 orders (20.01%) and Debit Card with 7,624 orders (15.25%). Wallet is the least-used method, with 1,947 orders (3.89%), showing that usage is concentrated more heavily in UPI and card-based payments than in wallets.

- **Business implication:** IndiaKart should maintain a smooth UPI and card payment experience while monitoring payment success and failure rates separately. This chart measures payment-method usage, not whether transactions made through each method succeed.

8. Age Distribution

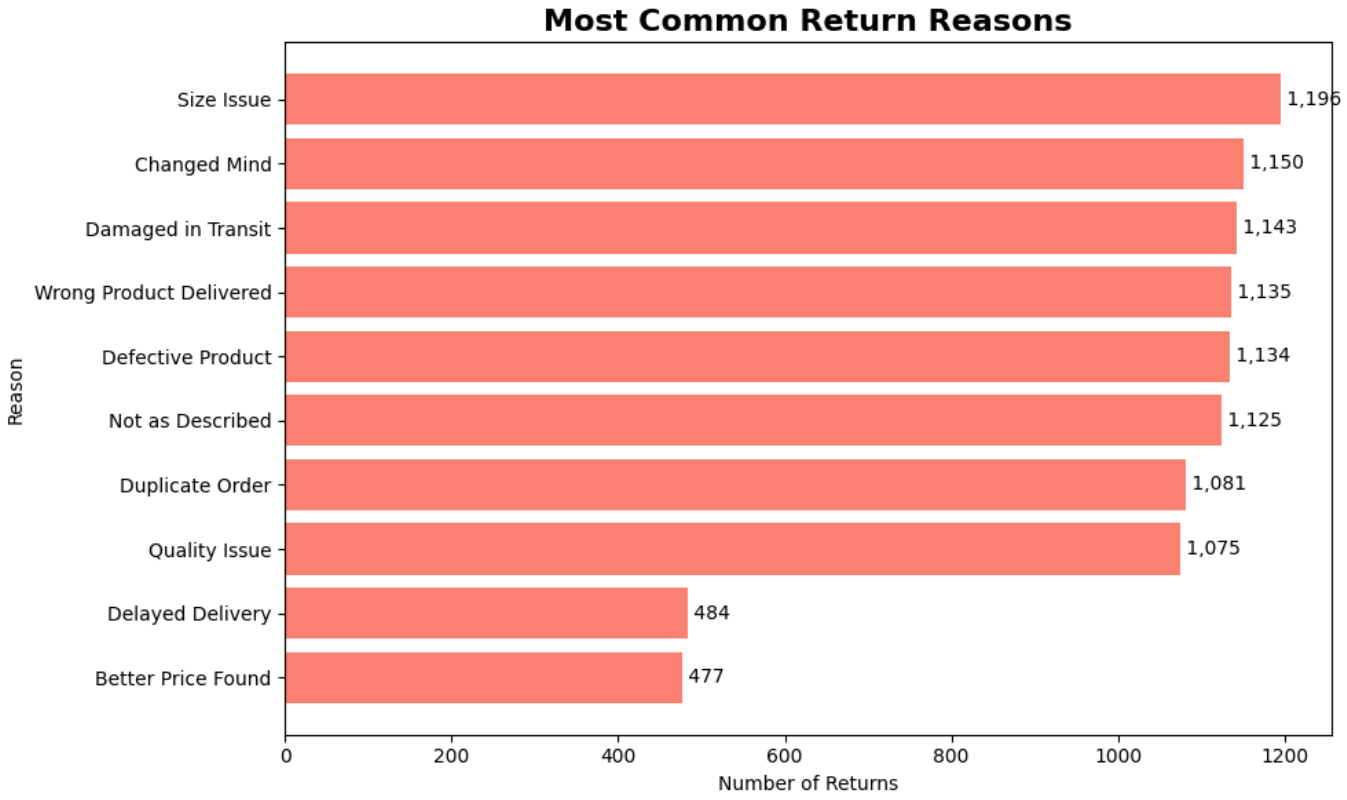


Business question: *What does the age profile of IndiaKart's customers look like?*

Customer ages range from 18 to 65, with an average age of about 41.7 years and a median of 42 years. The middle 50% of customers fall between 30 and 54 years of age, indicating that the customer base spans a broad adult age range rather than being concentrated exclusively among very young customers.

- **Business implication:** IndiaKart can use this age profile as a starting point for understanding its customer base, but it should combine age with purchase history, product preferences and customer segment before developing targeted marketing campaigns.

9. Return Reasons

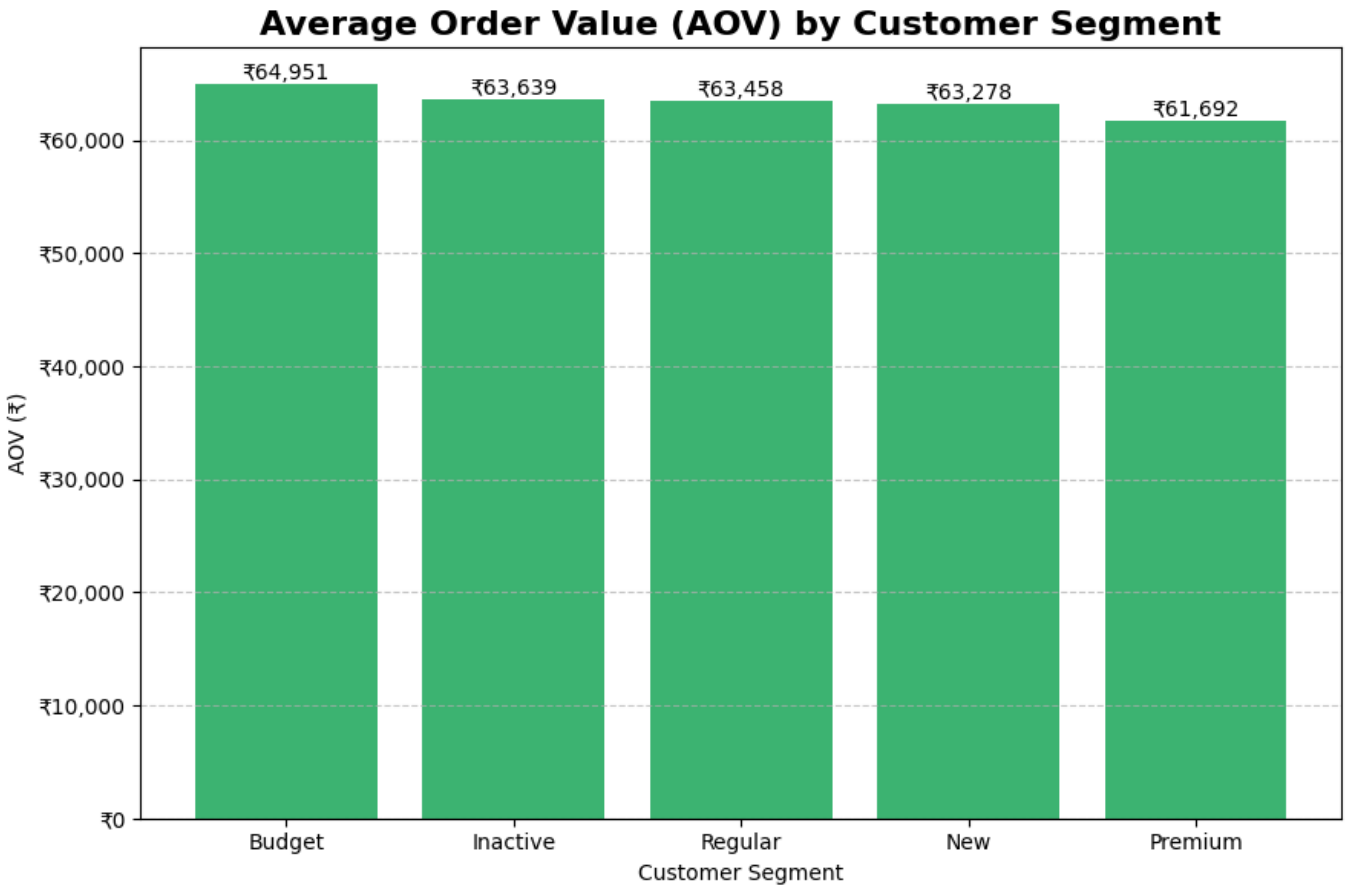


Business question: *What are the most common reasons customers return orders?*

Size Issue is the most common recorded return reason, with 1,196 returns (11.96%), followed by Changed Mind with 1,150 (11.50%) and Damaged in Transit with 1,143 (11.43%). Wrong Product Delivered, Defective Product and Not as Described are also common, whereas Delayed Delivery and Better Price Found have fewer records, at 484 and 477 respectively; overall, the return reasons are spread across several categories rather than being dominated by a single cause.

- **Business implication:** IndiaKart should investigate sizing information, product descriptions, quality checks and packaging or transit handling to identify preventable returns. Since the differences between several leading reasons are relatively small, the business should not focus exclusively on Size Issue without examining the others.

10. AOV by Customer Segment



Business question: *Which customer segments have the highest average value per delivered order?*

Budget customers have the highest calculated AOV at approximately ₹64,951, followed by Inactive customers at ₹63,639 and Regular customers at ₹63,458. Premium customers have the lowest AOV among the five segments at approximately ₹61,692, so the results do not support the assumption that Premium customers necessarily place the highest-value individual orders.

- **Business implication:** IndiaKart should investigate differences in product mix, discounts and purchasing patterns across segments before designing segment-specific offers. AOV measures average revenue per delivered order; it is not the same as total customer spending or customer lifetime value.